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KINEMATICS OF THE MARTAIGA BLOCK OF THE KUZNETSK-ALATAU PALEOISLAND ARC DURING THE LATE VENDIAN—EARLY ORDOVICIAN AS INFERRED FROM PALEOMAGNETIC DATA

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The Vendian-Early Cambrian, Middle Cambrian, and Early Ordovician volcanosedimentary deposits of the Martaiga block of the Kuznetsk-Alatau paleoisland arc (55.5° N, 88° E) have been investigated paleomagnetically. Their paleomagnetic poles have been determined, permitting us to plot the apparent polar wandering path (APWP) of this block from late Vendian through Early Ordovician. On the basis of this APWP and comparative positions of paleomagnetic poles of the Martaiga block and Siberian Platform, we succeeded in proving the kinematics of horizontal displacements of the Martaiga segment of the Kuznetsk-Alatau paleoarc relative to the Siberian continent. This movement is characterized by a southward drift from the equatorial latitudes of the Northern Hemisphere (about 10° N) to the tropical latitudes of the Southern Hemisphere (about 15° S); the velocity of the latitudinal component of movement reached its maximum in the Late Cambrian (about 6 cm/year) and gradually attenuated to the Early Ordovician (about 2 cm/year); drift of the block was complicated by the clockwise rotation for more than 50°; the amplitude of rotation was about 1°/myr and also attenuated from Cambrian to Ordovician. A considerable discrepancy between the Martaiga paleomagnetic poles and corresponding poles of the paleo-Siberian continent in the range from Vendian to Late Cambrian indicates that these tectonic units were separated in space. The collision of the Kuznetsk-Alatau paleoarc with continent occurred in the Early Ordovician; their paleomagnetic poles did not differ by that time. It has been established that the formation of the modern structural plan of the Kuznetsk Alatau was due to a large-scale displacement of structures along the periphery of the Siberian continent. The Martaiga block and other tectonic units of the Kuznetsk-Alatau paleoisland arc collided with the paleo-Siberian continent on the background of submeridional movements toward each other, with simultaneous clockwise rotation of the paleoarc.

Kuznetsk-Alatau paleoisland arc, paleomagnetic pole, apparent polar wandering path, kinematics of horizontal displacements

INTRODUCTION

Much attention has recently been given to problems of geodynamic and geotectonic conditions of formation of the folded framing of the Siberian Platform [1–7]. The paleoisland-arc complexes of the Altai-Sayan folded area are key objects permitting interpretation of the Early Paleozoic stage of development of the southwestern framing of Siberia. The paleomagnetic data obtained in recent years on island-arc complexes [3, 7–11] provided a considerable progress in understanding kinematic character of the development of the Siberian margin of the Paleoasian Ocean and permitted us to correct the paleogeodynamic reconstructions. At present, a safe paleomagnetic description in the Kuznetsk Alatau is available for the V-$Є_1$ and $Є_{2–3}$ complexes of the Zolotokitat block and

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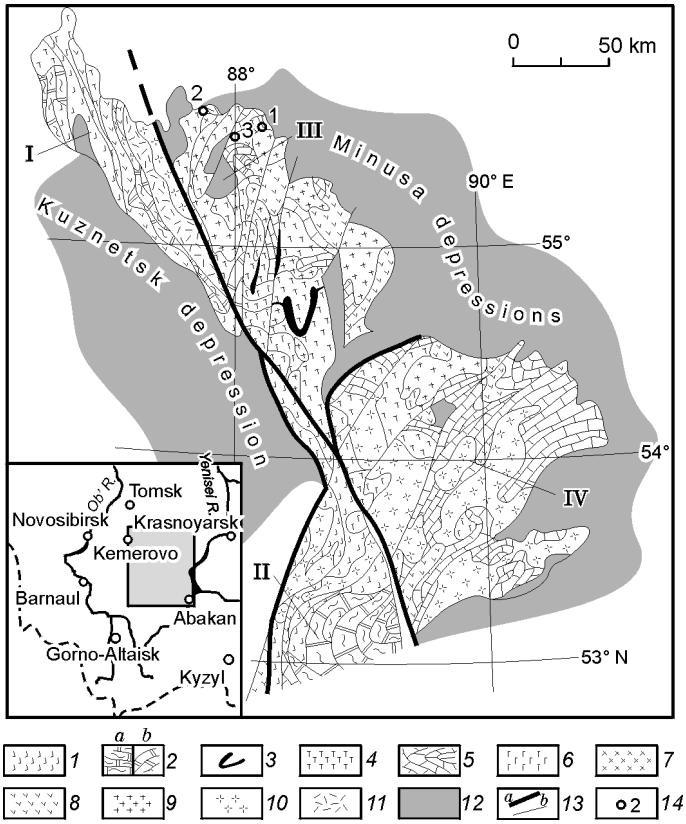


Fig. 1. Scheme of geological complexes of Kuznetsk Alatau. Complexes: *1*, *2* — of within-plate ocean islands: *1* — volcanogenic, *2* — carbonate: *a* — in structures of accretionary prism, *b* — in basement of island arc; *3–7* — undeveloped island arc *3* — ophiolite, *4* — volcanogenic, *5* — carbonate, *6*, *7* — predominantly intrusive: *6* — gabbroids, *7* — granitoids; *8*, *9* — developed island arc: *8* — volcanogenic, *9* — granitoid; *10*, *11* — formed by collision and strike slip: *10* — granitoid, *11* — volcanogenic; *12* — of Hercynian superimposed thrusts; *13* — faults separating (*a*) and complicating (*b*) large blocks; *14* — locations of studied sections: 1 — Bol'shoi Berchikul' (see Fig. 2), 2 — Chumai (see Fig. 3), 3 — Kiya (see Fig. 4); Roman numerals designate large blocks: I — Zolotokitat, II — Mras-Su, III — Martaiga, VI — Iuys-Batenev.

$Є_2$ complexes of the Iyus-Batenev block [8]. Despite a certain progress in paleomagnetic studies of the Early Paleozoic, the available data are not sufficient to restore a complete kinematic pattern of development of the region from origin of island arcs to their collision with the Siberian continent. To solve this problem is to construct apparent polar wandering paths (APWP) for each of the main tectonic blocks for some successive time sections. This paper reports first results of using this approach, exemplified by paleomagnetic studies of the Vendian-Lower Paleozoic formations of the Martaiga block of the Kuznetsk-Alatau paleoisland arc.

BRIEF GEOLOGICAL DESCRIPTION OF THE REGION AND OBJECTS OF PALEOMAGNETIC STUDIES

The Kuznetsk Alatau, occupying the central part of the Altai-Sayan folded area, is characterized by mosaic structure. In the generalized form, its structure is represented by four large tectonic blocks: Zolotokitat, Mras, Martaiga, and Iyus-Batenev (Fig. 1), which are fragments of the Kuznetsk-Alatau paleoisland arc, tectonically superimposed on the Siberian continent as a result of collision and subsequent strike-slip deformations. First two blocks are made up of complexes of the exterior of the paleo-island arc (accretionary prism and fore-arc trough). The Martaiga block is represented chiefly by complexes of magmatic paleoarc [12, 13]. The Iyus-Batenev block

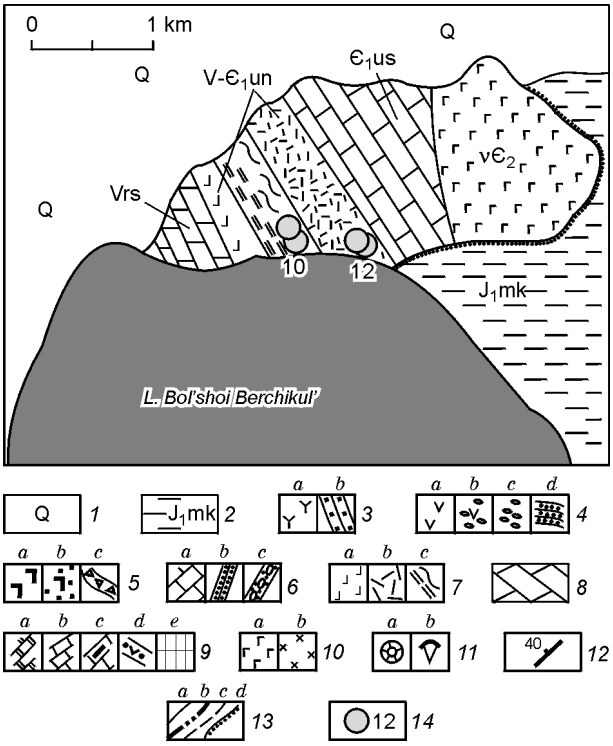


Fig. 2. Scheme of gelogical structure of lake Bol'shoi Berchikul' region (after data of geologic survey and thematic works of Zapsibgeologiya Enterprise). Deposits: *1* — Quaternary; *2* — Jurassic coaliferous of Makarov Formation (J_1mk); *3* — Devonian volcanogenic (*a*) and terrigenic (*b*) of Byskar Formation ($D_{1–2}$bs); *4* — Lower Ordovician andesite porphyries (*a*), tuff-conglomerates (*b*), conglomerates (*c*), and sandstones (*d*) of Smirnov Formation (O_1sm); *5* — Middle Cambrian basalt porphyries (*a*), their tuffs and tuffites (*b*), xenotuffs and clasto-lavas (*c*) of Berikul' Formation (Є$_2$) ; *6* — Lower Cambrian limestones (*a*), sandstones (*b*), and conglomerates (*c*) of Usinsk (Єus) and Karchit (Є$_1$kč) Formations; *7* — Vendian-Lower Cambrian basalt porphyries (*a*), porphyries (*b*), and clayey- and coal-siliceous schists (*c*) of Chumai (V-Є$_1$čm) and Ust'-Anzas (V-Є$_1$un) Formation; *8* — Vendian dolomites of Rastai Formation (Vrs); *9* — Upper Riphean marbles (*a*), limestones (*b*) with layers of coal- and clayey-siliceous schists (*c*), diabase porphyries and their tuffs (*d*), and siliceous rocks (*e*) of West Siberian Formation (R_3zs); *10* — intrusion rocks of basic (*a*) and neutral (*b*) composition; *11* — locations of finding of archeocyathids (*a*) and trilobites (*b*); *12* — elements of rock deposition; *13* — borders: geologic (*a*), tectonic in open (*b*) and closed (*c*) areas, *d* — with structural discordance; *14* — localities of paleomagnetic sampling.

is also mostly composed of complexes of magmatic paleoarc, which are replaced in its southeastern part by complexes of the inner slope of the paleoisland arc and then by deeper-water complexes of back-arc basin (southeastern Batenev Range and Azyr-Tal Ridge) [14–16]. Two stages are recognized in the development of paleoisland arc, represented by pertinent geological complexes [12, 13, 17].

The early-island-arc complex (see Fig. 1) consists of the Vendian-Lower Cambrian essentially volcanogenic formation comparable to the tholeiitic series of island arcs [12, 13]. Within the explored region, this complex is represented by the Ust'-Anzas and Chumai Formations, developed in the Kozhukhov and Martaiga structure-formational zones, respectively. According to Alabin [18], the Chumai Formation occurs in the north of the Martaiga block. The volcanic rocks of the Ust'-Anzas, which belong to a trachyte-trachybasalt-basalt association are characterized by cover-type effusions, higher saturation of sections in lavas with abundant pillow varieties, and few, if any, pyroclastic facies. As a rule, they are represented by basaltoids, which belong to a moderately alkaline series, are weakly differentiated, highly melanocratic, rich in oxides of Ti and P and depleted in Al. The lower age boundary of the formation is inferred from the concordant occurrence of volcanosedimentary formations on

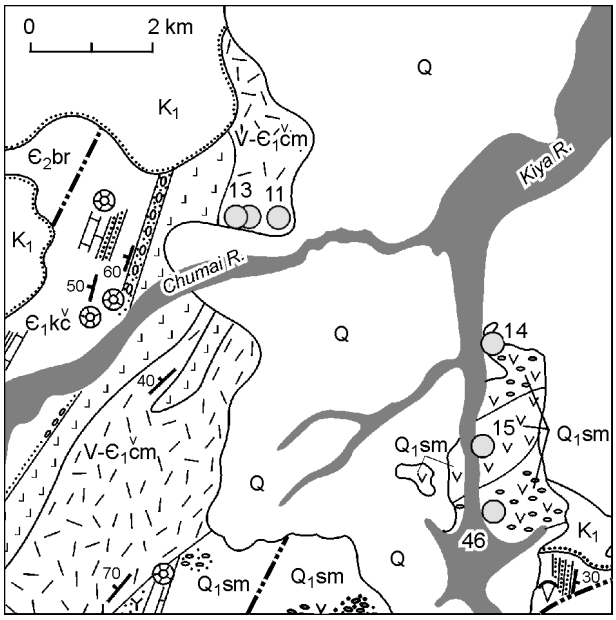


Fig. 3. Scheme of geological structure of Chumai Village region (after data of geologic surveys and thematic works of Zapsibgeologiya Enterprise). Designations follow Fig. 2.

the Vendian dolomites of the Rastai Formation [19]. From above, the Ust'-Anzas Formation is overlapped (with uncertain nature of contact) by archeocyatic limestones of the Usa Formation of the Early Cambrian Atdabanian Stage.

The volcanic rocks of the Chumai Formation, which belong to the plagiorhyolite-basalt association [19], are represented by basalt, andesite-basalt porphyries, tuffs of porphyries, quartz porphyries, lava breccias of porphyries, and tuff sandstones. They are characterized by low total alkalinity, abnormally low contents of K_2O, and low alumina content. The lower boundary of the formation is not established. The Chumai Formation is overlapped by concordant deposits of the Karacharov (Usa) Formation of the Early Cambrian Sanashtykgol Stage [18].

The complex of developed island arc is represented by Middle Cambrian successively differentiated series with well-expressed calc-alkaline and subalkaline composition and abundant pyroclastics [12, 13]. Typical representatives of this complex are volcanosedimentary deposits of the Berikul' Formation, which, according to Borisov [20], formed at the stratotype locality under conditions of a linear system of stratovolcanoes of the rear part of the island arc. The Berikul' volcanic rocks belong to a rhyodacite-basalt-trachybasalt association [19] and are represented by lavas, tuffs, and tuffites of moderately alkaline and, less frequently, calc-alkaline porphyry basalts as well as andesite-basalts, replacing one another from bottom to top of the section. With an angular unconformity, they overlap the archeocyate limestones of the Usa Formation. Near the bottom of the Berikul' Formation, there are typically xenotuffs and clastolavas with fragments of underlying limestones. The age of these formations is inferred from findings of skeletal fauna of Amgan age (middle and, possibly, upper Early Cambrian) in interbeds of tuffaceous siltstones from the lower section of the formation [21]. The upper boundary of the formation is marked by a regional gap; therefore, the upper age limit is not determined exactly.

Formation of the Ordovician deposits within the Kuznetsk Alatau is linked with the accretion-collision processes [16, 22–24], having led to the accumulation of thick molassa (Algain and Amzas Formations) and origin of volcanoplutonic complexes of higher alkalinity (volcanic rocks of the Taimen and Kitat Formations, gabbro-granitoid massifs of the Tebin, Karnayul', Upper Kondoma, and Tigertysh complexes) [19, 22]. In the explored section, the Ordovician is represented by the Smirnov Formation (analog of the Taimen Formation). Its composition is dominated by variegated conglomerates, sandstones, siltstones, marls, and finely layered limestones, alternating with tuff sandstones, andesite-basalt porphyries, porphyry tuffs of a trachydacite—trachyandesite-basalt association [19]. The above-mentioned deposits with a gap and angular unconformity rest on Cambrian formations and, also with an unconformity, are overlain by Devonian continental redstones. The time of formation of the Smirnov Formation corresponds to the Arenigian Stage of the general stratigraphic scale [25].

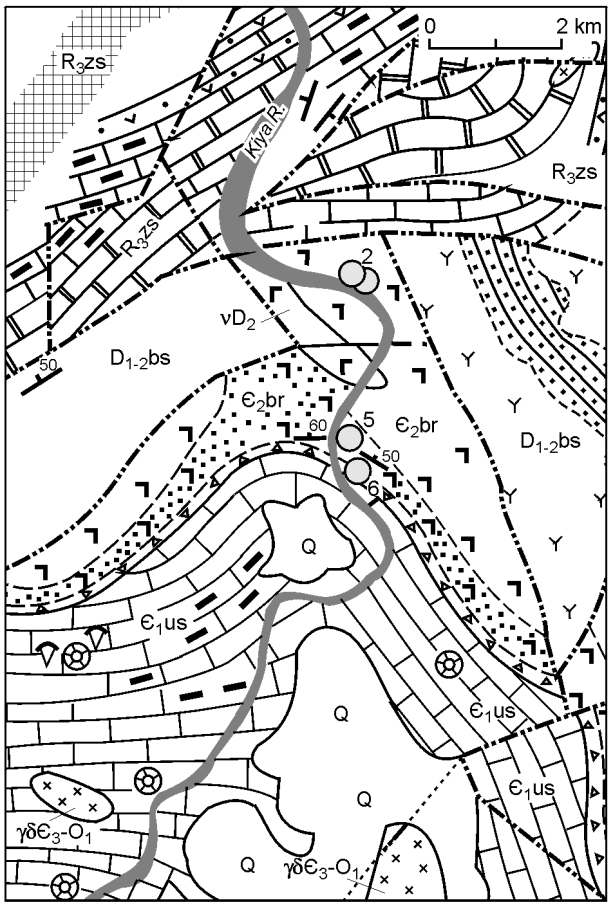


Fig. 4. Scheme of geological structure of Kiya River middle reaches region (after data of geologic surveys and thematic works of Zapsibgeologiya Enterprise). Designations follow Fig. 2.

Paleomagnetic studies in the northern part of the Martaiga block were carried out on volcanosedimentary formations which characterize three main stages of development of the Kuznetsk-Alatau paleoarc from origin in the late Vendian—early Cambrian (Ust'-Anzas Formation near Lake Bol'shoi Berchikul' (Fig. 2) and Chumai Formation on the northern margin of Chumai Village (Fig. 3) through the stage of developed island arc in the Middle Cambrian (Berikul' Formation in the middle reaches of the Kiya River upstream from Makaraksky Village (Fig. 4) to strike-slip-collision stage in the Early Ordovician (Smirnov Formation on the right bank of the Kiya River opposite Chumai Village (see Fig. 3). Average coordinates of the region under study are: 55.5° N and 88° E.

METHODOLOGY AND RESULTS OF PALEOMAGNETIC STUDIES

The laboratory experiments were carried out on the equipment of the Paleomagnetic Center of the Institute of Geophysics, Novosibirsk. Magnetic susceptibility was measured by a device made by K.S. Burakov of the United Institute for Physics of the Earth, Moscow. The direction of vector of natural remanent magnetization, NRM, was measured using a JR-4 spin-magnetometer (Czech Republic) and HSM cryogenic magnetometer (Germany) with a sensitivity of $1 \cdot 10^{-6}$ A/m. Stepwise thermal demagnetization to 700 °C at an interval of 30–40 °C was carried out using a shielded furnace, devised by V.P. Aparin (Institute of Physics, Krasnoyarsk). To carry out an Irs (*T*) thermomagnetic analysis, we used a thermorock-generator devised by K.S. Burakov. The obtained results were treated by standard algorithms with the use of component analysis [26], Halls' great circles method [27], or combined analysis [28], in the form of computer programs [29]. Analysis of the obtained paleomagnetic poles and calculation of kinematic parameters of plate motion was performed using the GMap program [30].

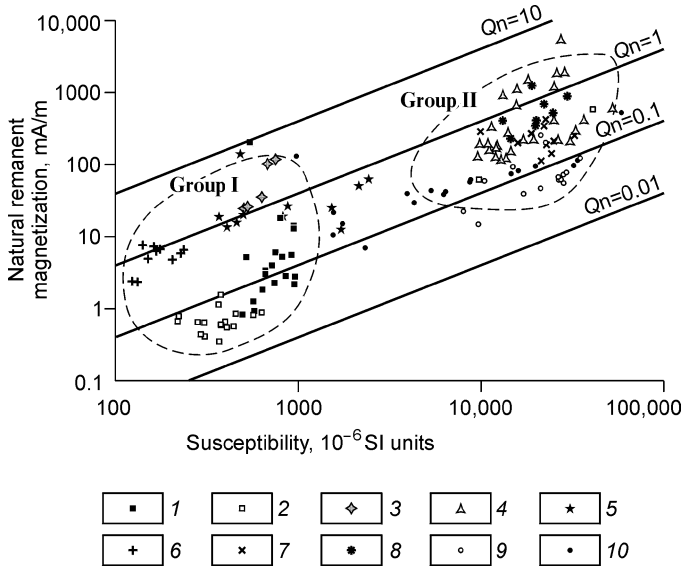


Fig. 5. Scalar magnetic characteristics of studied rocks. *1*, *2* — porphyries of Ust'-Anzas Formation (V-$Є_1$): *1* — points Ki-10 and Ki-10a, *2* — Ki-12a and Ki-12b; *3*, *4* — andesite-basalt porphyries of Chumai Formation (V-$Є_1$): *3* — Ki-11, *4* — Ki-13a and Ki-13b; *5–7* — volcanites of Berikul' Formation ($Є_2$): *5* — bottom of formation, tuffs, Ki-5, *6* — bottom of formation, clastolavas, Ki-6, *7* — top of formation, basalt porphyries, Ki-2; *8* — dike of diabase porphyries in basalts of Berikul' Formation, Ki-2d; *9*, *10* — volcanogenic-deposit rocks of Smirnov Formation (O_1): *9* — tuff-conglomerates, Ki-14, *10* — andesite porphyries, Ki-15.

According to values of magnetic susceptibility and value of natural remanent magnetization, the rocks can be subdivided into two groups occupying quite compact areas on a logarithmic diagram of NRM/*X* (Fig. 5). The group of weakly magnetic varieties, with $X = 100–1000 \cdot 10^{-6}$ SI units, NRM = 0.5–50 mA/m, includes volcanosedimentary rocks of the Ust'-Anzas Formation (points Ki-10, 10a, Ki-12a, 12b), tuffs and tufflavas of the bottom of the Berikul' Formation (points Ki-5 and Ki-6) as well as basalts of the Chumai Formation (point Ki-11). The group of magnetic varieties includes the extrusive rocks of the Chumai Formation (points Ki-13a, 13b), porphyries from the top of the Berikul' Formation (point Ki-2), diabases of the dike breaking through these rocks (point Ki-2d), as well as volcanosedimentary deposits of the Smirnov Formation (points Ki-14, Ki-15). The magnetic susceptibility for this group is characterized by values of $5000–40{,}000 \cdot 10^{-6}$ SI units and NRM varies from 50 to 2000 mA/m.

The composition of magnetic minerals is quite diverse in the explored rocks. The unblocking temperatures obtained by Irs (*T*) thermomagnetic [31] and stepwise thermal demagnetization of NRM suggest that magnetite, Ti-magnetite, Ti-maghemite, and pyrrhotite may occur there.

Magnetite is inferred from Irs and NRM unblocking temperatures, T_{ub} = 575 °C (Fig. 6, *a*). Inflections on the curve of primary heating within the range 250–350 °C (see Fig. 6, *a*) may indicate the presence of maghemite. Magnetite is supposed to be the main carrier of magnetization for rocks of the Ust'-Anzas and Berikul' Formations.

Titanomagnetite and titanomaghemite are identified from a drastic drop of Irs and NRM magnetizations in the temperature range 350–400 °C, which corresponds to Curie temperatures of the magnetite-ulvospinel series with TiO_2 content x = 0.15–0.25 [32]. These bends are established in rock specimens of the Chumai and Ust'-Anzas Formations. In addition to the bending, only T_{ub} of magnetite is recorded in the specimens. Differences between supposed Ti-magnetite and Ti-maghemite are well visible on comparison of curves of reheating. Titanomagnetite is characterized by higher Irs on remagnetization of specimen and shift of bending point (unblocking temperature) toward the region of higher temperatures (see Fig. 6, *b*), which is a diagnostic feature of Ti-magnetite [31]. For

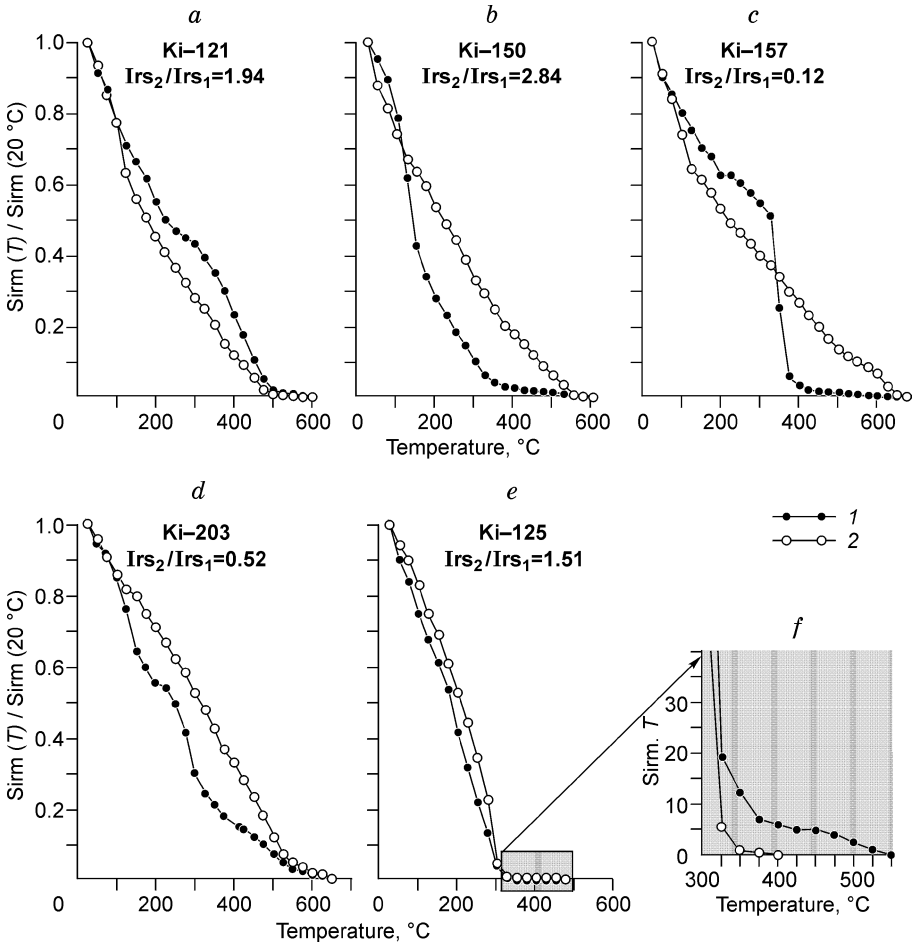


Fig. 6. Characteristic curves Irs (*T*). *a–e* — From thermomagnetic analysis, *f* — enlarged fragment of high-temperature range for specimen Ki-125. Heating: 1 — first, 2 — second. Notes in text.

Ti-maghemite, on the contrary, the value of magnetization after heating decreases, and the curve bending is not recorded on reheating in the range corresponding to unblocking temperature. The reason is that Ti-maghemite transforms to hematite on the first heating (see Fig. 6, *c*).

Hematite is identified from unblocking temperature of NRM and Irs magnetizations, T_{ub} = 675 °C (see Fig. 6, *d*). It is typical only for volcanosedimentary rocks of the Smirnov Formation and is not established in other explored stratigraphic divisions. In the studied specimens, it is associated with magnetite and, possibly, with Ti-magnetite, which is inferred from Irs (*T*) curve bending in corresponding temperature intervals.

Pyrrhotite. Indications of this mineral are unblocking temperatures of 325–350 °C (see Fig. 6, *e*) and a specific increase in magnetization after the first heating, related to a new magnetic phase with a blocking temperature of 480–500 °C (see Fig. 6, *f*). Pyrrhotite is supposed to occur only in specimens of weakly magnetic basalts of the Chumai Formation (point Ki-11).

Three main groups of magnetization components have been established in NRM of studied rocks: low-temperature, medium-temperature (group **B1-B2**), and high-temperature components (group **A1-A4**).

The low-temperature component is decomposed on heating to 200 °C and, certainly, is a viscous magnetization, acquired in laboratory. This component is excluded from further consideration. Distribution of the medium- and high-temperature components among objects of study is of more complicated nature.

Late Vendian—Early Cambrian. Magnetization of the Ust'-Anzas rocks is made of two components. The medium-temperature component (**B1**) is destroyed at 320 °C. The predominant polarity of this component is reverse (in sign of declination), and distribution of its directions is of specific "banana" shape. It is possible that **B1** is a superposition of two or more components with overlapped unblocking temperatures. In some specimens, component

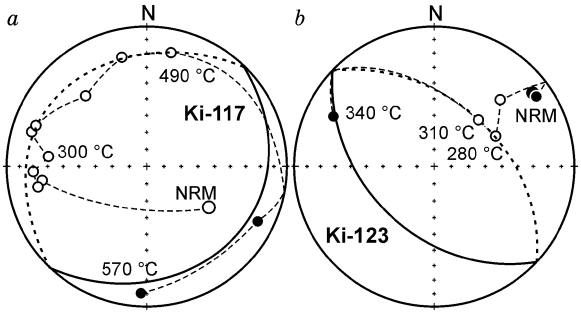


Fig. 7. Typical stereoplots of thermal demagnetization of rocks of Ust'-Anzas (*a*) and Chumai (*b*) Formations. The example of plotting of great circles in temperature ranges related to basic magnetization-bearing minerals (notes in text). Stratigraphic coordinates.

A1 is recorded in the range 320–480 °C. However, because of the overlapped spectra of blocking temperatures, direction **A1** is usually recognized only by the great circles method plotted for the range 320 to 570 °C (Fig. 7, *a*).

Destruction of the medium-temperature component in the rocks of the Chumai Formation also occur below 300 °C. However, unlike the Ust'-Anzas Formation, this component (component **B2**) has normal polarity. The direction corresponding to component **A1** is recorded in the Chumai Formation at $T > 320$ °C. As earlier, to determine it, we used the great method circles. In the Chumai basalts, where pyrrhotite carry magnetization (site Ki-11), the range of existence of component **B2** (270–300 °C) is considerably overlapped by the interval of unblocking temperatures of component **A1** (200–340 °C). Therefore, to determine the direction of **A1**, great circles are plotted for a range from 200 to 340 °C, i.e., to the Curie point for pyrrhotite (see Fig. 7, *b*).

Middle Cambrian. In the volcanosedimentary rocks of the Berikul' Formation and intruding dike of diabases, the medium-temperature component is destructed at temperatures below 320–360 °C. Its direction is irregular but in general corresponds to the directions of components **B1** and **B2** of the Ust'-Anzas and Chumai Formations. Above this temperature limit, the magnetization vector on the Zijderveld diagram goes toward the origin. As a rule, starting from 510–540 °C, the linear segments indicating the destruction of characteristic component of magnetization, ChRM, are well identifiable. A complete demagnetization occurs at $T = 590$ °C. The direction of ChRM in the rocks of the Berikul' Formation regularly changes throughout the section. For the lower part of the formation, component **A2** is determined, which is characterized, in stratigraphic coordinates, by gentle inclinations, about 0 °C (Fig. 8, *a*). The rocks of the top section of the Berikul' Formation is characterized by component **A3**. Its directions differ from **A2** by steeper inclination (~20 °C) at the same values of declination (see Fig. 8, *b*). The dike cutting the Berikul' deposits contains component **A4**. Unlike **A2** and **A3**, it has the normal polarity by inclination sign and is distinguished by steeper values of inclination (see Fig. 8, *c*).

Lower Ordovician. The character of thermal demagnetization of rocks of the Smirnov Formation is quite monotypic. Nearly all specimens have a component of reverse polarity, which is interpreted as **B1**. The upper temperature limit of the existence of this component varies from 400 to 550 °C. In some cases, at higher temperatures, a component is recorded, which is close in direction to component **A4**, but is usually antipodal to it. Because of considerable superposition of spectra of blocking temperatures of these components, the direction of **A4** can be established with less errors, using the method of great circles. Typical stereoplots indicating permanent destruction of magnetization components along the arc of great circle are shown in Fig. 9.

DISCUSSION

Components **B1** and **B2** have secondary superimposed character, which follows from their distribution over the explored objects: on the one hand, both components are present in objects of different ages and, on the other hand, component **B1** occurs in objects covering a wide range, from Early Cambrian to Early Ordovician. The presence of stable components **B** in an average interval of temperatures is an argument for the fact that secondary magnetization has a partial thermoremanent or thermoviscous rather than viscous nature, which implies partial heating of rocks after formation of the primary component. Potential secondary alterations of rocks are also inferred from low values of *Qn* (see Fig. 5), which are typical of postmagmatically altered igneous rocks [33]. The

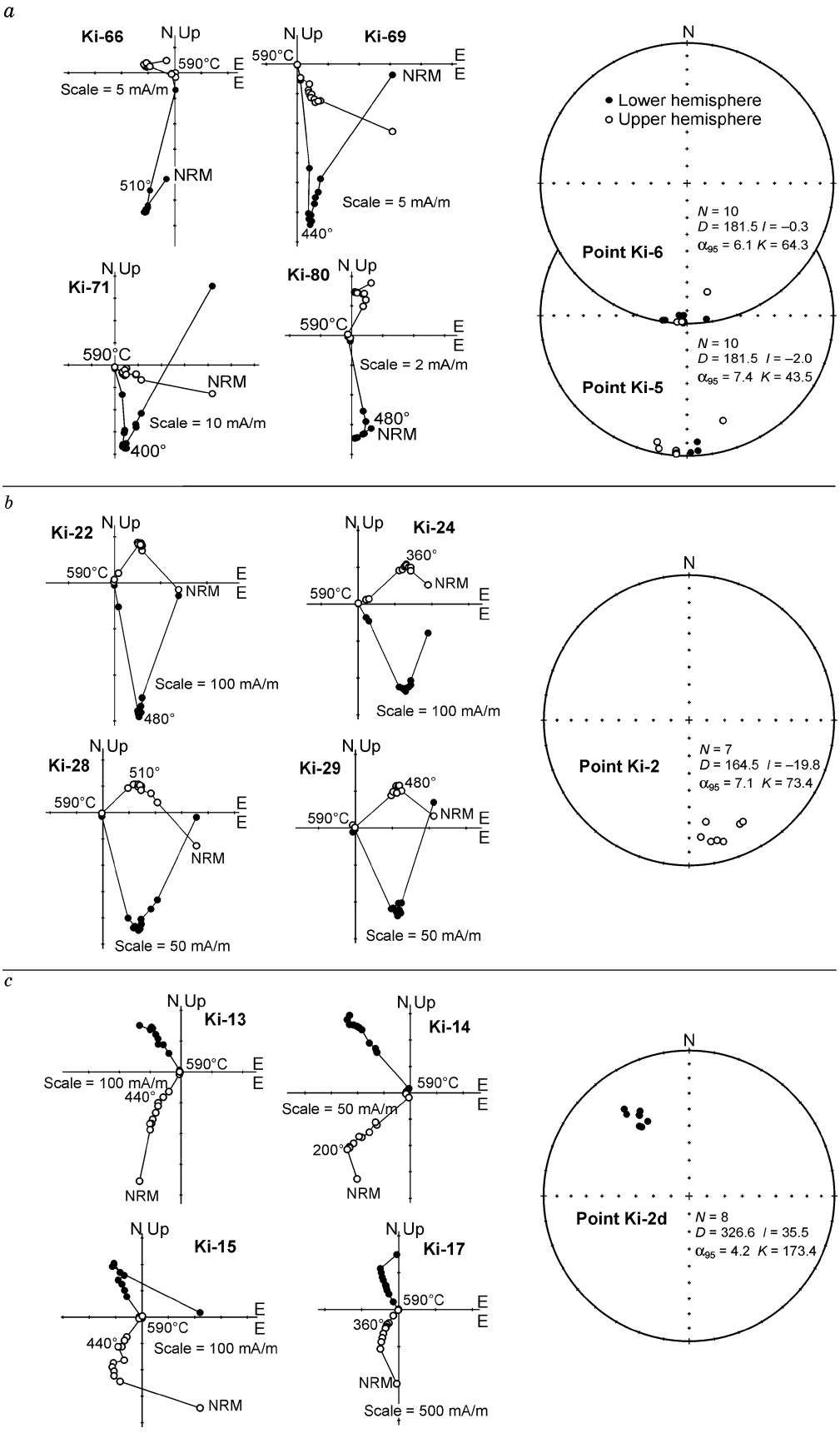


Fig. 8. Characteristic Zijderveld diagrams and ChRM components distribution: *a* — for rocks from bottom of Berikul' Formation section; *b* — for rocks from top of Berikul' Formation section; *c* — for dike of diabase porphyries.

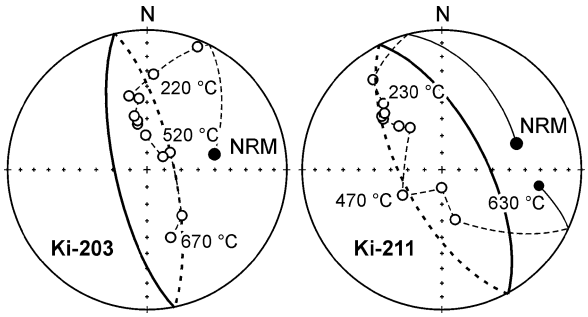


Fig. 9. Typical stereoplots for thermal demagnetization of rocks from Smirnov Formation. An example of plotting of great circles, stratigraphic coordinates.

mechanism of formation of secondary magnetization can be different. For example, for rocks of the Chumai and Ust'-Anzas Formations the secondary component may be due to low-temperature oxidation of Ti-magnetite, whereas for rocks of the Smirnov Formation, to formation of secondary hematite. The main argument for the secondary nature of components of **B1** and **B2** is time of their formation relative to time of folding. Negative fold test (in a modification of Watson-Enkin [34], at a level of means of direction sampling sites) suggests that the formation of these components occurred after folding or corresponds to its final stage. The maximum value of precision parameter is achieved at 10% of defolding (Fig. 10). The paleomagnetic pole calculated from directions of

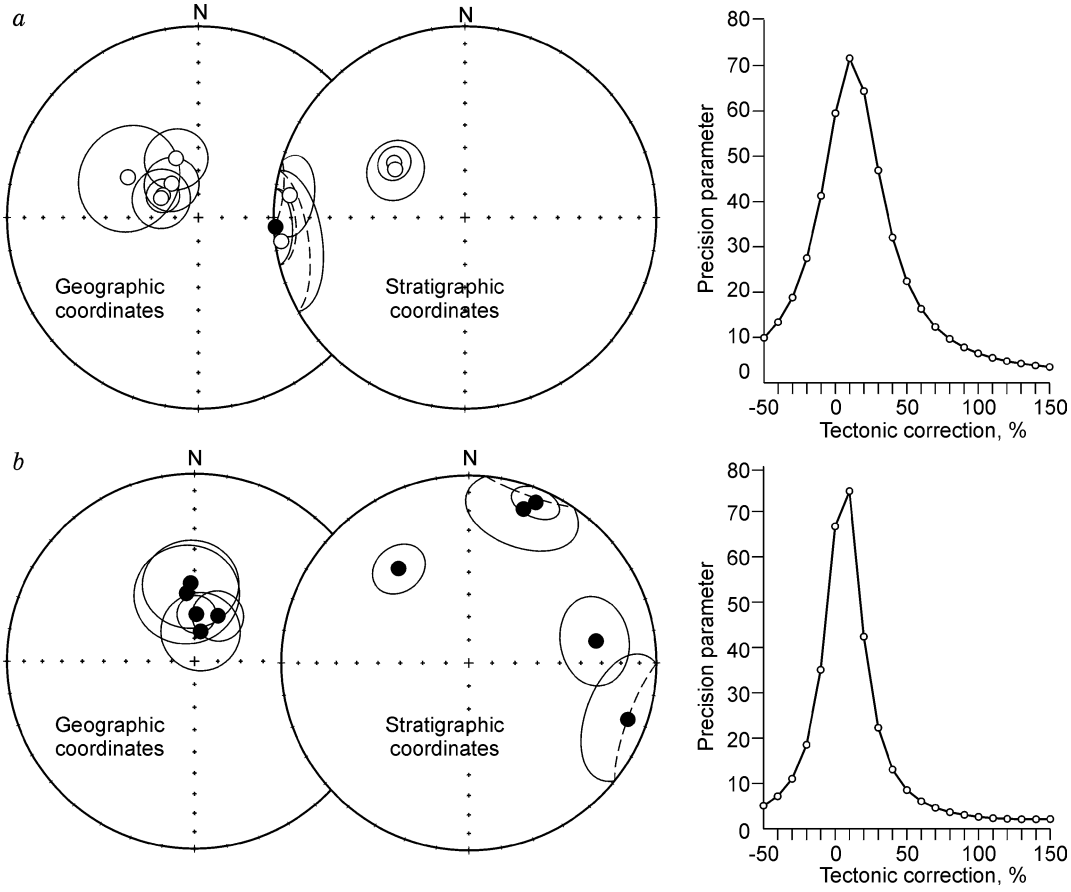


Fig. 10. Results of fold test [34] for stable magnetization directions averaged over sampling localities: *a* — component B1, *b* — component B2.

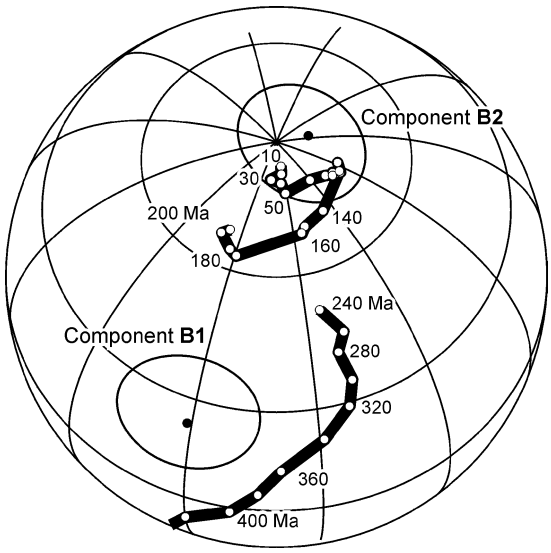


Fig. 11. Paleomagnetic pole positions from components B1 and B2 as compared with APWP of Siberia [35] and Eurasia [36].

component **B1** (in the geographic system of coordinates) lies near the Upper Paleozoic segment of the Siberian APWP [35], thus suggesting that this component formed in the Carboniferous (Fig. 11). For component **B2** the pole within a confidence limit does not differ from the Cenozoic interval of the Eurasian APWP [36]; therefore, the age of this component is considered the Cenozoic (see Fig. 11).

Directions of **A1-A4** seem to be the closest to directions of primary magnetization. Prefolding character of magnetization is proved by fold tests for components **A1** (Early Cambrian), **A2** (lower Middle Cambrian), and **A4** (Early Ordovician) (Fig. 12). The obtained data show that paleomagnetic directions in the dike cutting the Berikul' deposits completely correspond to the Lower Ordovician. Additional support for the primary origin of components **A** comes from temperature limits of their existence, which in all cases correspond to blocking temperatures of main magnetization-carrying minerals.

Given the directions in the Smirnov Formation with negative inclination characterize normal polarity (as assumed for the Early Ordovician of the Siberian Platform [37]), the principle of minimal displacements of paleomagnetic pole requires negatively inclined directions to be taken as normal polarity for the Middle Cambrian, and positively inclined directions, as normal polarity for the Vendian–Early Cambrian. The latter agrees with results of magnetostratigraphic studies of the Lower Cambrian of the Siberian Platform, where the deposits of the Tommotian Stage have chiefly normal polarity by sign of inclination [38]. Based on this interpretation, the Vendian-Early Cambrian, Middle Cambrian, and Early Ordovician paleomagnetic poles form quite a regular trend (Fig. 13), an apparent polar wandering path for the Martaiga block. The main characteristics of this APWP are as follows: (1) on the background of considerable rotation of plate clockwise during the Cambrian (approximately in the range from 540 to 480 Ma), the pole had a sublatitudinal drift of southeastern direction; (2) at the Vendian-Cambrian boundary it was near the eastern margin of Africa at the latitudes of Madagascar and then moved toward the southwestern coast of Australia, to more moderate latitudes, about 40 °S; (3) over the considered period the pole wandering gradually becomes slower; (4) the Early Ordovician pole of the Martaiga block differs little from the coeval pole of the Siberian Platform [35]. Since that time, the kinematics of the movement of the Martaiga paleomagnetic pole seems to follow that of the APWP of the Siberian Platform.

Given the revealed regularity in the position of poles, component **A3**, recognized for the top section of the Berikul' Formation seems to characterize the late Middle or, possibly, early Late Cambrian. The paleomagnetic pole of **A3** well fits the APWP of the Martaiga block, with its age assumed to be 505–500 Ma.

The plotted APWP permits calculation of the main kinematic parameters of horizontal movements of the Martaiga block in the range from Late Vendian-Cambrian to Early Ordovician (Fig. 14). We can state that, from Late Vendian through Early Ordovician, the Martaiga block moved from north to south, from region of equatorial (~10 °N) to more moderate tropical latitudes (~15 °S), approaching the Siberian continent, whose northern (in old coordinates) margin was at 0–10 °N [35]. Given the adopted age of components, the rate of this movement did

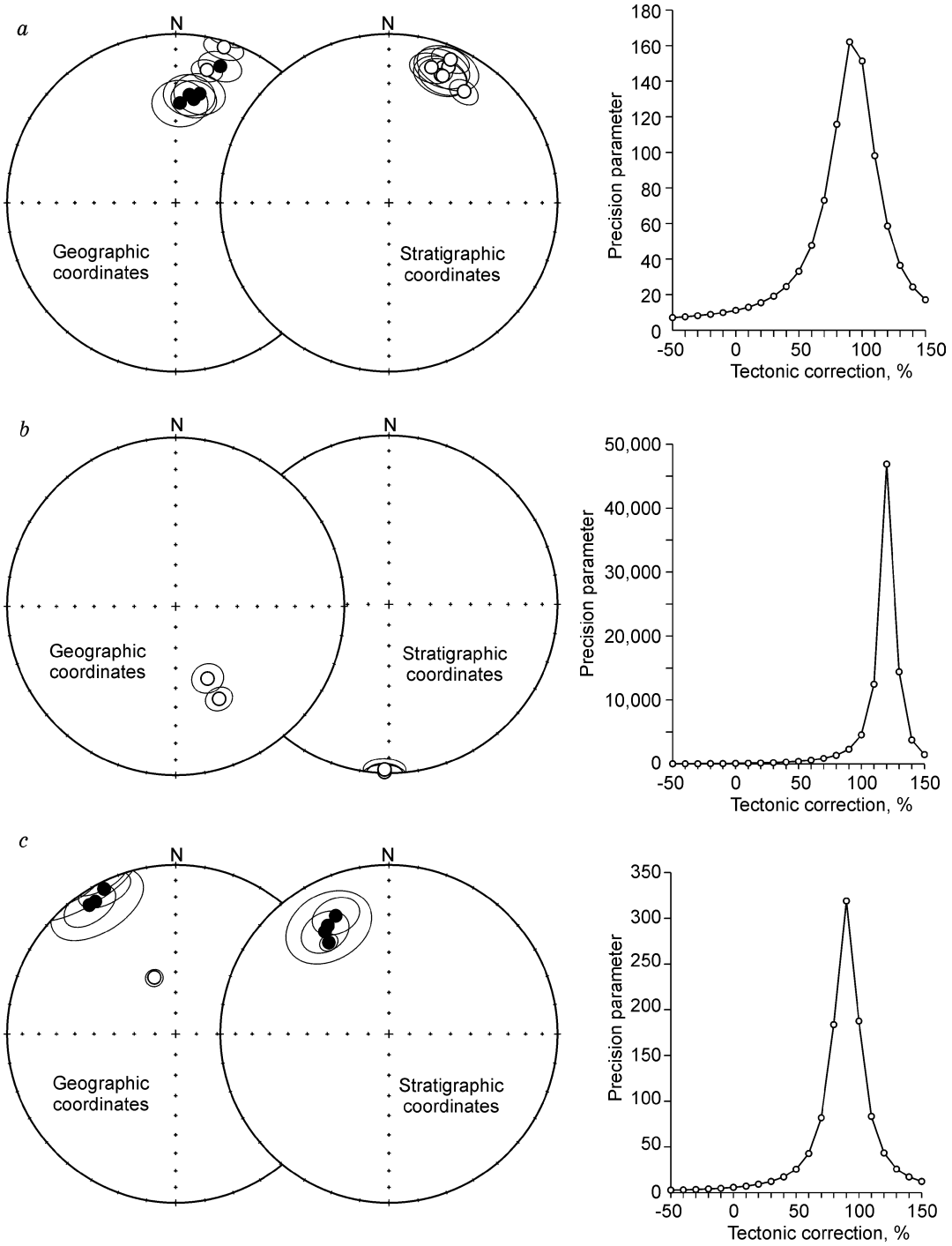


Fig. 12. Results of fold test [34] for stable magnetization directions averaged over sampling localities: *a* — component A1, *b* — component A2, *c* — component A4.

not exceed 0.5°/myr, which is commensurate with the rate of movement of the Siberian continent at that time [35]. A gradual change of paleomagnetic inclination shows that the southern drift of the Martaiga block was complicated by a clockwise rotation for 50° or more. As the block moved, both latitudinal and angular components of velocity gradually attenuated. The maximum of displacement in latitude (along longitude) falls on the Early–early Middle Cambrian (see Fig. 14). At that time, the meridional drift rate was 5–6 cm/year. At the same time, the block

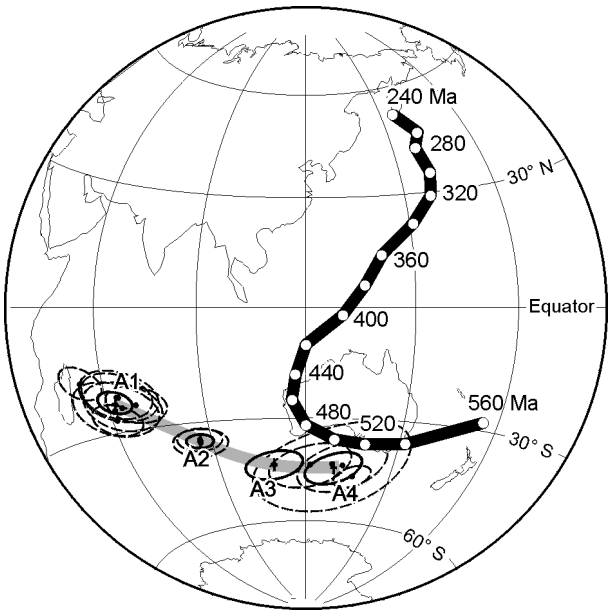


Fig. 13. Paleomagnetic pole positions of studied objects from components A1-A5 as compared with APWP of Siberia [35].

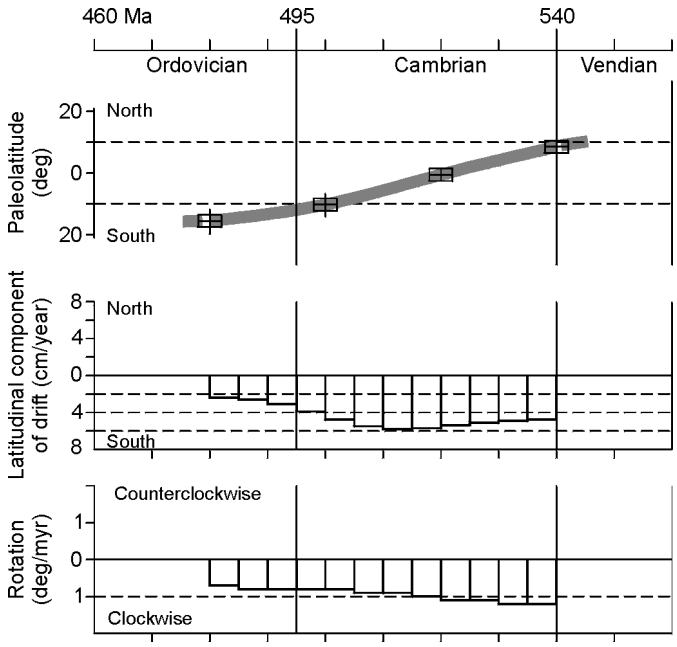


Fig. 14. Kinematic parameters of movement of Martaiga Block of Kuznetsk Alatau paleoisland arc from Late Vendian through Early Ordovician.

rotation was also maximal, somewhat more than 1°/myr. Since the late Cambrian and in Early Ordovician, the rate parameters were no more than 2–3 cm/year and about 0.7°/myr, respectively.

Comparison of apparent polar wandering paths of the Siberian continent and Martaiga block (see Fig. 13) evidently shows that these structures were spatially separated in the Vendian–Cambrian, and their horizontal movements were directed oppositely. At that time, the APWP's of Siberia and Martaiga block differed drastically,

Table 1

Paleomagnetic Results from Cambrian-Ordovician Complexes of Martaiga Block (55.5° N, 88° E)

Sampling locality	Component	*n/N*	D_g	I_g	K_g	$\alpha_{95}g$	D_s	I_s	K_s	$\alpha_{95}s$
Vendian-Early Cambrian										
Ust'-Anzas Formation										
Ki-10	A1	1z+7c/10	7.2	35.8	32.3	11.1	21.6	–15.3	32.3	11.1
Ki-10a	A1	7c/9	12.4	34.3	45.0	11.0	22.2	–19.8	45.1	11.0
Ki-10 and Ki-10a	B1	13z/15	339.1	–63.2	10.8	13.2	277.1	–8.5	10.8	3.2
Ki-12a	A1	1z+6c/8	2.4	40.6	32.2	12.3	23.8	–13.4	32.2	12.3
Ki-12b	A1	11c/11	9.9	37.7	26.4	9.9	23.1	–19.8	26.4	9.9
Ki-12a and Ki-12b	B1	8z/19	321.1	–71.8	24.0	11.5	267.2	1.0	24.0	11.5
Chumai Formation										
Ki-11	A1	5c/5	18.1	16.1	171.7	8.6	17.5	–17.4	174.5	8.5
Ki-11	B2	4z/5	11.7	76.8	30.0	17.1	80.2	31.8	30.0	17.1
Ki-13a	A1	9c/9	17.3	–4.4	64.9	7.3	23.4	–8.8	65.1	7.3
Ki-13b	A1	2z+9c/18	13.0	–20.7	60.2	6.3	34.2	–21.8	60.1	6.3
Ki-13a and Ki-13b	B2	15z/27	27.2	67.8	13.0	11.0	323.3	37.3	13.0	11.0
Middle Cambrian										
Berikul' Formation										
bottom, Ki-5	A2	10z/11	156.2	–51.3	43.5	7.4	181.5	–2.0	43.5	7.4
bottom, Ki-5	B2	8z/11	2.1	69.6	45.9	8.3	22.6	8.1	45.9	8.3
bottom, Ki-6	A2	10z/10	154.9	–39.2	64.2	6.1	181.5	–0.3	64.3	6.1
bottom, Ki-6	B1	4z/10	299.3	–55.2	18.3	22.1	262.4	–3.5	18.2	22.1
top, Ki-2	A3	7z/10	137.7	–35.0	73.3	7.1	164.5	–19.8	73.4	7.1
top, Ki-2	B2	5z/10	357.1	55.7	15.6	20.0	19.5	14.0	15.7	20.0
dike, Ki-2d	A4	8z/8	339.0	–61.0	173.6	4.2	326.6	35.5	173.4	4.2
dike, Ki-2d	B2	3z/8	353.2	60.1	31.8	22.2	109.6	11.0	31.8	22.2
Early Ordovician										
Smirnov Formation										
Ki-14	A4	12c/18	333.7	4.7	22.7	10.0	335.7	24.3	2.7	10.0
Ki-15	A4	12c/19	326.1	9.0	20.5	10.6	327.9	29.3	20.5	10.6
Ki-14 and Ki-15	B1	23z/37	301.5	–72.3	19.3	7.1	307.4	–51.5	19.3	7.1
Ki-46	A4	6c/7	328.7	9.1	22.0	18.8	330.5	27.2	22.0	18.8
Ki-46	B1	5z/7	296.7	–72.2	38.6	12.5	304.2	–53.6	38.6	12.5

Note. *n/N* — number of samples statistically processed (z — PCA direction, c — great circle) relative to the total of samples; *D* — declination in degrees, *I* — inclination in degrees, α_{95} — 95%-confidence cone, *K* — precision parameter; indices "g" and "s" are geographic and stratigraphic coordinate systems, correspondingly.

which indicates that the Siberian continent moved northward on the background of counterclock rotation. Some coincidence of the poles is observed only in the Ordovician. This suggests the tectonic superposition of structures of the Martaiga block and Siberian continent at the indicated time, which is completely confirmed by the available geological data [17, 39]. It is at that time (at the Late Cambrian–Ordovician boundary) that the rotation sense changed for the Siberian continent, which can also be evidence of considerable tectonic rearrangement of plate movements [35], related to collisional processes.

Table 2

Mean Directions of Natural Remanent Magnetization, Corresponding Paleomagnetic Poles, and Paleolatitudes of Block (55.5° N, 88° E)

Component	*n*/*N*	D_g	I_g	$K_g/\alpha_{95}g$	D_s	I_s	$K_s/\alpha_{95}s$	Pol	Φ	Λ	dp/dm	Plat
A1 (V?-$Є_1$)	7/58	12.0	20.4	11.2/18.9	23.6	–16.7	151.4/4.9	R	–23.0	62.5	2.6/5.1	8.5 ± 3.6
A2 ($Є_1$?-$Є_2$)	2/20	155.5	–45.3	89.3/26.7	181.5	–1.1	4546.7/3.7	N	–35.0	86.2	1.9/3.7	–0.6 ± 2.6
A2* ($Є_1$?-$Є_2$)	2/20	155.5	–45.2	41.5/5.1	181.5	–1.1	54.5/4.5	N	–35.0	86.2	2.3/4.5	–0.6 ± 3.2
A3* ($Є_2$–$Є_3$?)	1/7	137.7	–35.0	73.3/7.1	164.5	–19.8	73.4/7.1	N	–43.1	109.1	3.9/7.4	–10.2 ± 5.4
A4 (O_1)	4/38	330.8	–7.9	5.8/42.0	330.3	29.1	187.7/6.7	R	–44.0	129.6	4.1/7.4	–15.6 ± 5.5
B1 (C?)	5/50	411.8	–67.8	59.5/10.0	279.4	–23.9	6.5/32.5	R	23.6	118.9	14.0/16.7	50.8 ± 15.3
B2 (N?)	5/35	4.2	66.5	66.7/9.4	41.1	29.4	2.6/59.2	N	83.0	244.8	12.7/15.5	49.0 ± 14.0

Note. Directions of components of natural remanent magnetization averaged over localities, starred — over samples; *n*/*N* — number of localities and number of samples statistically processed; *D* — declination, *I* — inclination in degrees; *K* — precision parameter; α_{95} — radius of 95%-confidence oval; indices "g" and "s" are modern and ancient coordinate systems, correspondingly; Pol — polarity: N — normal, R — reverse; Φ and Λ — geographical coordinates of paleomagnetic pole, longitude and latitude, correspondingly; dp/dm — semiaxes of 95%-confidence ellipse for paleomagnetic pole; Plat — paleolatitude in degrees: positive values for Northern Hemisphere, negative — for Southern Hemisphere.

The available paleogeodynamic and paleotectonic reconstructions [6, 7] suggest that the kinematics of horizontal movements established for the Martaiga block is typical of two other tectonic units of the Kuznetsk Alatau: Zolotokitat and Iyus-Batenev blocks. A similar character of horizontal movements is also recorded for the Gorny Altai segment of a paleoisland-arc system developed on the margin of the Siberian continent in the Early Paleozoic, with the only difference that the velocities and scale of movements might be somewhat different.

Thus, the history of formation of the modern structural plan of the Kuznetsk Alatau is chiefly related to large-scale movements of blocks along the periphery of the Siberian continent in the Early Paleozoic. The Martaiga block and other tectonic units of the Kuznetsk Alatau collided with the Siberian continent on the background of counter submeridional movements and simultaneous clockwise rotation of the Kuznetsk-Alatau paleoarc.

CONCLUSIONS

1. Paleomagnetic studies have shown that from Late Vendian through Early Ordovician the Martaiga block of the Kuznetsk-Alatau paleoisland arc moved from north to south, from equatorial latitudes to more moderate tropical regions at a velocity of no more than 0.5 °/myr. The southerly drift of the Martaiga block was complicated by a clockwise rotation for more than 50°. As the block moved, both latitudinal and angular components of the velocity gradually attenuated.

2. A discrepancy in path of apparent wandering of paleomagnetic poles of the Siberian continent and Martaiga blocks indicates that in the Vendian-Cambrian these structures were spatially separated, and intersection of both APWP's in the Early Ordovician suggests a tectonic superposition of this structures at that time.

3. Formation of the modern structural plan of the Kuznetsk Alatau is caused by large-scale displacements of structures over the periphery of the Siberian continent in the Early Paleozoic. The accretion of the Martaiga block and other tectonic units of the Kuznetsk Alatau to the continent occurred on the background of counter submeridional movements and simultaneous clockwise rotation of the Kuznetsk-Alatau paleoarc.

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